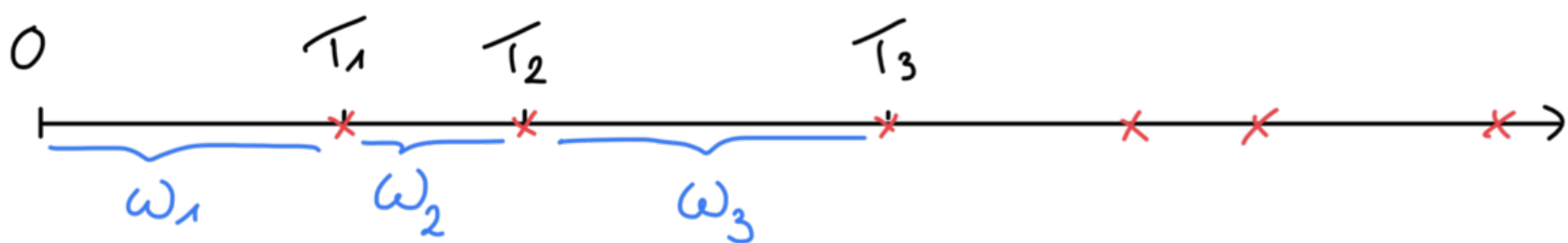


EXPONENTIAL DISTRIBUTION & POISSON PROCESS

WE HAVE A STRING OF EVENTS, LIKE TRADES



① WAITING TIMES ω_j INDEPENDENT

$$\omega_j \sim \text{exponential}(\lambda)$$

$\lambda \rightarrow$ rate of events ($\lambda = 5 \text{ trades/sec}$)

② # OF EVENTS IN TIME t IS POISSON (λt)

$$\frac{\lambda^k e^{-\lambda k}}{k!}$$

Ex: Trades are observed at a rate of 5/sec ($\lambda = 5$)

$\Rightarrow$ Probability I wait t secs. for next trade

$$P(T > t) = e^{-5t}$$

$\Rightarrow$ Probability no trades in next t secs.

$$P(k=0 \text{ in } t \text{ secs}) = e^{-5t}$$

$\Rightarrow$ Probability 1 trade in next t secs

$$P(k=1 \text{ in } t \text{ secs}) = 5e^{-5t}$$

GAMMA DISTRIBUTION

Waiting time for the n^{th} arrival:

$$T_r = \omega_1 + \omega_2 + \dots + \omega_r ; T_r = \text{gamma}(r, \lambda)$$

$$P(T_r > t) = P(N_t \leq r-1) = \sum_{k=0}^{r-1} \frac{e^{-\lambda t} (\lambda t)^k}{k!}$$

$$f(t) = e^{-\lambda t} \frac{\lambda^r \cdot t^{r-1}}{(r-1)!}$$

PDF of gamma
✓

Ex: We have 100 Mln \$ USD/EUR long position.

To hedge risk we want to off-load the position.

We decide to slice risk and wait 3 distinct buying orders ($r=3$).

If orders come at an average of $\lambda = 2/\text{min}$

$\rightarrow$ Estimated time to clear the risk:

$$E(T) = \frac{r}{\lambda} = \frac{3}{2} = 1.5 \text{ min}$$